

GridPulse 330: DevOps Power Grid Telemetry & Alert Console

Executive-level Technical System Architecture and Data Flow



1. METEOROLOGICAL RADAR & FORECAST DATA INGEST

LEVEL 1

- Convective storm tracking, lightning density, ambient temp, wind shear.

(Anticipated Weather State)

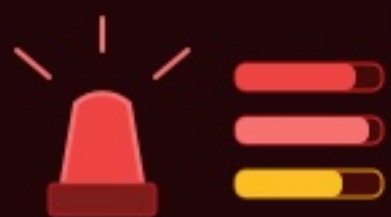


2. GRID SCADA TELEMETRY & PHYSICAL MODEL

LEVEL 2

- 330 kV Bulk Transmission Corridors (Double Circuit Lines 1 & 2)
- System Bus Voltage (Nominal 330.0 kV, Statutory: 313.5 - 340.0 kV)
- System Frequency (Nominal 50.00 Hz, Statutory: 49.80 - 50.20 Hz)
- Active / Reactive Power Balance (Generation MW vs. Load Demand MW, MVAR)

(Telemetry Stream)

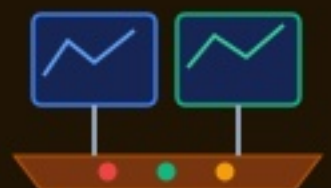


3. THRESHOLD & CONTINGENCY RULE ENGINE

LEVEL 3

- **Rule V-HIGH:** If Voltage ≥ 340.0 kV $\rightarrow$ High Overvoltage Alarm (Ferranti Risk)
- **Rule F-HIGH:** If Frequency ≥ 51.00 Hz $\rightarrow$ High Frequency Alarm (Load Rejection)
- **Rule N-1 TRIP:** Lightning trip on Line 1 $\rightarrow$ Parallel Line 2 Overload Warning ⚠

(Alarm & Advisory Dispatch)



4. OPERATOR REMEDIATION & DISPATCH ACTION

LEVEL 4

- **Action A:** Switch IN 330 kV 150 MVAR Shunt Reactor (suppress voltage to ~ 333 kV)
- **Action B:** Fast Turbine Governor De-load (-400 MW, restore 50.00 Hz frequency)
- **Action C:** Post-deionization Synchro-check auto-reclose on Line 1
- Sequence of Events (SOE) timestamped audit logging



Welcome to GridPulse 330

Designed & Engineered by [Samuel Peter](#) • Critical Infrastructure DevOps



WHAT THIS APP DOES

GridPulse 330 is an interactive DevOps Power Grid Telemetry & Alert Console engineered for National Control Centers. It models the acute physical impacts of rapid, unpredictable weather shifts on **330 kV transmission lines** and **50.00 Hz grid frequency**, alerting operators before blackouts occur.

✓ MAIN SYSTEM CAPABILITIES

- **Meteorological Contingency Radar:** Tracks Clear/Sunny, Cloud Inversion, Gale Gusts, and convective Thunderstorms with heavy rainfall and lightning shear in real-time.
- **Statutory Dual-Threshold Alarms:** Instant acoustic & visual sirens when 330 kV Bus Voltage exceeds **>340.0 kV** (Ferranti effect) or Frequency breaches **>51.00 Hz** (load rejection).
- **Cascading Line Protection:** Real-time monitoring of dual-circuit 330 kV bulk lines (Ikeja West – Oshogbo) with ANSI 87L differential protection and thermal N-1 transfers.
- **Operator Dispatch Action Playbook:** Closed-loop remedial dispatch controls to switch in 150 MVAR Shunt Reactors, fast-deload generation (-400 MW), and auto-reclose.

→ QUICK-START OPERATIONAL PLAYBOOK

- 1 Under **Anticipated Weather**, select **"Thunderstorm & Heavy Rain"** and trigger event.
- 2 Observe the **>340.0 kV High Voltage** and **>51.00 Hz High Frequency** sirens fire.
- 3 Use the **Operator Console** below to switch in the Shunt Reactor & De-Load Generation!

ENERGY SECTOR DEVOPS & SCADA EMS ARCHITECTURE

Adheres to Nigerian 330 kV Grid Code statutory operating ceilings and IEEE 1547 / IEC 60870 standards.



GRIDPULSE 330

DEVOPS POWER GRID SCADA & CONTINGENCY CONSOLE

50.02 Hz • STABLE

330.2 kV • NOMINAL

SIREN ON

S. Peter (Lead)

METEOROLOGICAL CONTINGENCY MONITORING DOPPLER RADAR LIVE

Simulate sudden atmospheric front shifts along the 330 kV transmission corridor (Clear Sunny → Convective Thunderstorm).

Clear / Sunny

Ambient: 34°C

Wind: 12 km/h • Light

STABLE

Cloud Inversion

Ambient: 26°C

Wind: 22 km/h • PV Drop

MODERATE

Gale Gusts

Ambient: 23°C

Wind: 68 km/h • Oscillation

CAUTION

Thunderstorm & Heavy Rain

Severe Lightning • Torrential Rain (45 mm/h)

Acute Differential Trip & Ferranti Surge Risk

CRITICAL CONTINGENCY

⚡ TRIGGER NEXT-MOMENT EVENT

Simulate Severe Storm & Line 1 Fault

330 kV BUS VOLTAGE (SUBSTATION BUS A)

330.2 kV

NOMINAL (SAFE)

300 kV 330 kV (Nominal) 340 kV (MAX)

Ferranti Guardrail: >340.0 kV triggers Shunt Reactor alert.

SYNCHRONOUS GRID FREQUENCY

50.02 Hz

SYNCHRONIZED

49.0 Hz 50.00 Hz (Target) 51.00 Hz (MAX)

Governor Droop: 4.0% • Rotor Inertia Constant H: 4.5 s

GENERATION VS DEMAND BALANCE

Total Grid Generation

4,820 MW

Substation Demand

4,818 MW

Active Balance ΔP : +2.0 MW (Equilibrium)

Reactive Flow Q: +115.4 MVAR (Inductive)

Spinning Reserve: +450 MW (Ready)

330 kV Bulk Line 1 (Ikeja West – Oshogbo Corridor)

ENERGIZED

Current Load

712 A

Power Transfer

407 MW

Thermal Capacity

49% (Safe)

330 kV Bulk Line 2 (Ikeja West – Oshogbo Corridor)

ENERGIZED

Current Load

708 A

Power Transfer

405 MW

Thermal Capacity

48% (Safe)

OPERATOR DISPATCH REMEDIAL ACTION PLAYBOOK (CLOSED-LOOP RESTORATION)

SCADA INTERLOCK: ALL REMEDIAL SEQUENCES ARMED

Switch IN 330kV Shunt Reactor [150 MVAR]

Mitigates Ferranti Open-Circuit Overvoltage (>340 kV)

Dispatch Fast Ramp-Down [-400 MW]

Arrests Severe Turbine Rotor Overspeed (>51.00 Hz)

Initiate Line 1 Auto-Reclose Sequence

Synchrocheck verification & dual corridor restoration

SCADA SEQUENCE OF EVENTS (SOE) AUDIT LOG

BUFFER HEALTHY

16:42:01.104 INFO Doppler radar reports convective front approaching Oshogbo corridor at 42 km/h.

16:42:04.288 TELEMETRY 330 kV Bus Voltage nominal at 330.2 kV; Synchronous frequency steady at 50.02 Hz.

16:42:07.892 SCADA Circuits 1 & 2 balanced; Line differential relays (ANSI 87L) armed with zero active alarms.